Comparing Weather Trends in State College and Philadelphia, 2006-2025

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1 Introduction

Weather influences a surprising number of everyday decisions, whether it's what to wear, planning events, or even what you make for dinner. Yet most people rarely look further than the next day's forecast. Two cities can be only a couple hundred miles apart in the same state and still feel like completely different climates to someone who has been to both. State College and Philadelphia are a great example of this. State College sits inland in the Appalachian foothills with an elevation of approximately 1,200 feet, while Philadelphia lies on the eastern coastal plain near sea level. Their geographic separation, along with the difference in elevation, raises a question: how different is the weather that these two cities actually experience?

This report will explore that question using the last two decades of daily weather records collected by the National Oceanic and Atmospheric Administration. There are over 7,000 daily observations for both cities, providing us with enough data to look at long-term weather trends and how their climates compare over the years.

Our work follows the Exploratory Data Analysis approach, focusing mainly on what the data actually reveals rather than trying to prove a specific point. We narrowed our analysis down to three main angles: how precipitation totals compare month by month, how temperature is distributed across the four seasons, and how each city's annual snowfall compares. Together, these angles paint a picture of two cities that are close in proximity but experience noticeably different climates.

The remainder of this report describes the data we used and its provenance, takes a closer look at each of our angles of comparison, and concludes with a discussion of the patterns we noticed through this analysis.

For this project, we are using a tidyverse data analysis perspective and a exploratory analysis perspective.

1.1 Paradigm and perspective

For the coding work, we use the tidyverse paradigm because the data is being cleaned, organized, summarized, and visualized step by step. Functions like mutate(), select(), group_by(), summarize(), and ggplot() allowed us to take the raw NOAA weather data and turn it into

readable tables and plots. This approach is useful because each step builds logically on the previous one, making the code easier to follow and explain.

For the data analysis, we use a exploratory perspective because our goal is to explore NOAA temperature data and identify patterns instead of testing one formal hypothesis. By summarizing temperatures by season and decade and creating visualizations, we can compare Philadelphia and State College and look for possible differences in temperature patterns over time. EDA is useful for this project because it helps reveal trends, seasonal changes, and location-based differences in the data.

2 FAIR and CARE Principles

FAIR Findable The data was fairly easy to locate. Publicly available on the National Oceanic and Atmospheric Administration's (NOAA) website with an entire section dedicated to explaining how to select the correct dataset. Being that a lot of weather apps and companies use NOAA's data and them being government agency, they almost encourage those to utilize the data for whatever personal project people want to do.

Accessible Having an entire webpage dedicated to selecting what dataset you want from any location and time frame in the United States, the data being inaccessible was not an issue. Again, NOAA is encouraging those to utilize their data.

Interperable NOAA has their own code for some weather conditions, such as using 1 or 0 if there was or wasn't fog in the area, respectively and just having confusing column naming systems, such as WT11 for high wind for example. A legend could be found on NOAA's website luckily, so when the csv files were uploaded into R Studio, we just had to rename the columns to make it easier to analyze the data later on when we got to the analysis portion of the code.

Reusable The data is absolutely reusable. Previous weather data is never going to change. Once the day passes, the weather is logged and stored for future reuse in data analytics. It's important to keep an organized weather data collection that is maintained often. NOAA does a great job with this to ensure future reusability when needed.

CARE Collective Benefit Weather data is not privately owned information, which is why NOAA offer the data to the public. Weather can be especially destructive at different times of the year; the data helps track potential or eminent floods, hurricanes, tornadoes, blizzards and various other natural disasters to help warn the community and keep the aware. Climate research is also done using NOAA data to help understand long term climate trends and see how we can protect the planet and make it as habitable as possible for life on Earth.

Authority to Control Being that NOAA is a government agency, their work is supposed to act in the best interest of the people, which we believe they have been. The data being free to the public is one example of this, however other weather companies such as The Weather

Channel and AccuWeather collect their own data in-house which is only available to them. They have been working to make it so NOAA can't publish their data for free as they feel it rivals their business model. Many scientists belive that weather has no borders however, and have been working to keep the weather data as open-source and collaborator friendly as possible. ## FAIR and CARE Principles # FAIR and CARE Principles FAIR Findable The data was fairly easy to locate. Publicly available on the National Oceanic and Atmospheric Administration's (NOAA) website with an entire section dedicated to explaining how to select the correct dataset. Being that a lot of weather apps and companies use NOAA's data and them being government agency, they almost encourage those to utilize the data for whatever personal project people want to do.

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Responsibility NOAA holds complete control over the accuracy of the data. In the grand scheme of things, being off by one degree on the temperature for one day seems like an insignificant difference overall, however when we have constantly advancing technology to track data, we want to assume that the data is being properly collected. It's also imperative when

we are studying something serious like the possibility of a natural disaster occuring. Now their is little we can do against a Category 5 Hurricane or extreme flooding in a short period of time, however we can be prepared for it. If the data is misread or collected poorly and we underestimate how strong a storm ends up being, than NOAA would have to take responsibility for misreporting the data and underpreparing the community.

Ethics

3 Data

3.1 Data Provenance

The data for this project comes from NOAA, the National Oceanic and Atmospheric Administration. NOAA collects daily weather data from weather stations across the United States, including stations near Philadelphia and State College. These stations record lots of information such as daily high temperature, low temperature, observed temperature, precipitation, snowfall, and snow depth.

For this project, we use two NOAA datasets: one for Philadelphia and one for State College. Each dataset contains daily weather observations from 2006 to 2025. We focus mostly on temperature variables, especially daily maximum temperature, minimum temperature, and average temperature. The data was originally collected to track local weather conditions and build long-term climate records.

We use this data to compare temperature patterns between Philadelphia and State College. Because both datasets come from NOAA and include the same types of measurements, they can be compared by season and by 10-year period.

3.2 FAIR and CARE Principles

3.2.1 FAIR

3.2.1.1 Findable

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3.2.1.2 Accessible

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3.2.1.3 Interperable

NOAA has their own code for some weather conditions, such as using 1 or 0 if there was or wasn't fog in the area, respectively and just having confusing column naming systems, such as WT11 for high wind for example. A legend could be found on NOAA's website luckily, so when the csv files were uploaded into R Studio, we just had to rename the columns to make it easier to analyze the data later on when we got to the analysis portion of the code.

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3.2.2 CARE

3.2.2.1 Collective Benefit

Weather data is not privately owned information, which is why NOAA offer the data to the public. Weather can be especially destructive at different times of the year; the data helps track potential or eminent floods, hurricanes, tornadoes, blizzards and various other natural disasters to help warn the community and keep the aware. Climate research is also done using NOAA data to help understand long term climate trends and see how we can protect the planet and make it as habitable as possible for life on Earth.

3.2.2.2 Authority to Control

Being that NOAA is a government agency, their work is supposed to act in the best interest of the people, which we believe they have been. The data being free to the public is one example of this, however other weather companies such as The Weather Channel and AccuWeather collect their own data in-house which is only available to them. They have been working to make it so NOAA can't publish their data for free as they feel it rivals their business model. Many scientists belive that weather has no borders however, and have been working to keep the weather data as open-source and collaborator friendly as possible.

3.2.2.3 Responsibility

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3.2.2.4 Ethics

NOAA operates different from a traditional company as they are government owned, meaning they give their data away vs some of the other aforementioned companies. weather is an important part of day to day activities for any one person. It impacts everyone. Making sure their is proper funding to support publicly available data in underrepresented communities and lower-income areas is imperative. This data should not flex to incomes, or skin color. It's numbers that can help save people and give everyone a chance to prepare for the day.

3.3 Attributes Used in This Analysis

4 Snowfall Comparison

Annual snowfall is one of the more visible differences you can see between the two cities, since both elevation and proximity to the coast affect how much snow a location gets every year. Figure 1 compares the yearly totals for both stations over the last two decades.

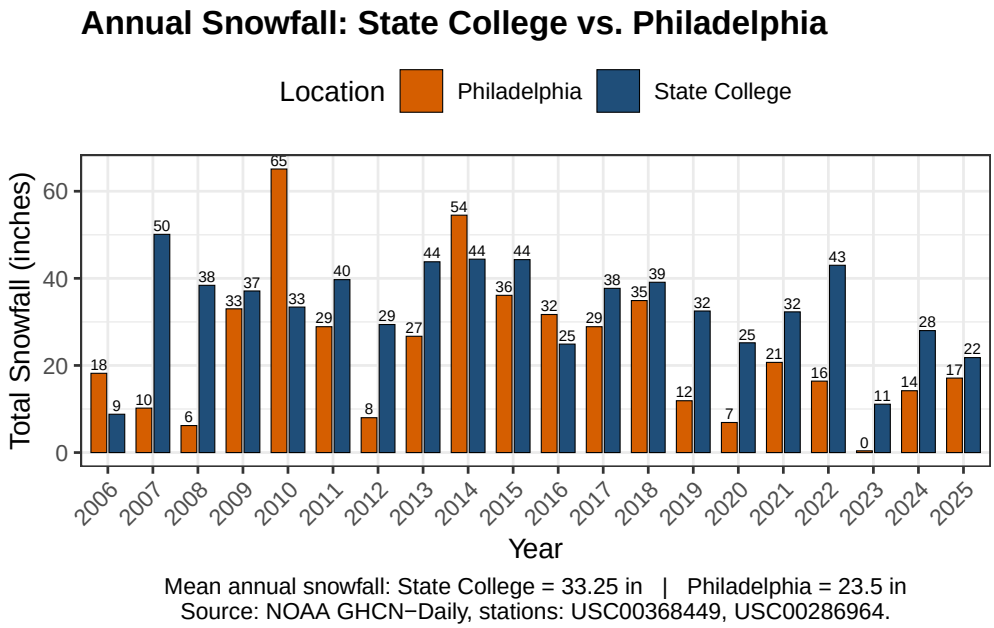

Figure 1: Annual snowfall totals for State College and Philadelphia, 2006-2025

This figure shows twenty grouped bar pairs, with one pair for each year between 2006 and 2025. In each pair, the navy bar shows State College's total snowfall for that year, and the orange bar shows the same total for Philadelphia. Each bar's height is the year's total snowfall in inches, and the value printed above gives the exact total. One of the first patterns Figure 1 reveals is that State College almost always sees a higher annual snowfall total than Philadelphia. State College's bar is taller than Philadelphia's in all but 4 years over this twenty-year span, and its average annual total is nearly 10 inches higher than Philadelphia's over that same period. This chart also reveals that Philadelphia has a much higher variance between the years than State College. Philadelphia experienced several years of single-digit snowfall, but in 2010 saw roughly three times its average annual total.

Together, these patterns suggest that both elevation and inland geography influence how much snow a city sees over the course of a winter. State College sits at a significantly higher elevation in the higher Appalachian terrain, which seems to give it consistently more snow than Philadelphia's lower coastal plain location. A lot of Philadelphia's variability is shaped by nor'easters. These coastal storms can bring huge amounts of snow if lined up correctly but can also miss the city completely.

While Figure 1 shows the year-by-year totals for each city, Table 1 shows that same twenty year window as a table of each city's summary statistics, including measures of variability, and the number of snow days at each station.

Table 1: Annual snowfall summary statistics for State College and Philadelphia, 2006-2025.

City	Mean(in)	Snowiest Year(in)	Lowest Year(in)	Std Dev	Snow Days/Year	Inches/Snow Day
Philadelphia	23.5	65.1	0.4	16.4	11.7	2.01
State College	33.2	50.1	8.8	10.9	26.6	1.25

The values in Table 1 support the trends that were displayed in Figure 1 while also providing more details and concrete numbers for analysis. In the grouped bar chart you could see that Philadelphia had more variability between years. With this table you can now see the exact measure of variability for both cities, with Philadelphia having roughly a 50% higher standard deviation. The most insightful portion of the table is the specific snow day statistics. State College has over double the snow days every year, but Philadelphia averages nearly twice as much total snowfall for each day it does snow. At first these two statistics might feel like they contradict each other, but they actually fully support the patterns shown by the plot. Philadelphia consistently has lower snowfall frequency, but when the nor'easter storms do happen to line up with the city, they bring huge amounts of snow down at one time. State College's location and higher elevation leads to more consistent snowfall throughout the year, and shows that its higher average comes from how frequently it snows there.

5 Temperature Distribution by Season

5.1 Introduction to temperature

6 Temperature Insights

This section explores the temperature data for Philadelphia and State College. The data includes daily maximum, minimum, and observed temperatures. We are trying to find out how temperature patterns compare between the two locations, how those patterns vary across the four seasons, and how the trends have changed or stayed the same over the last 20 years. This will be shown through a table and plots.

Table 2: Philadelphia Temperature Summary by Season and 10-Year Period

Decade	Season	Avg High	Avg Low	Avg Temp	Hottest	Coldest	Days
06-15							
06-15	Winter	42.6	25.2	33.9	73	-2	901
06-15	Spring	63.5	41.7	52.6	95	3	920
06-15	Summer	84.4	64.0	74.2	103	45	918
06-15	Fall	66.0	46.1	56.0	95	18	909

16-25							
16-25	Winter	44.1	25.8	35.0	77	-1	903
16-25	Spring	63.3	42.3	52.8	94	13	920
16-25	Summer	84.9	64.7	74.8	100	44	920
16-25	Fall	67.4	46.4	56.9	97	14	910

6.1 Philadelphia table interpretation

This table showcases the four seasons and data about them for both decades in degrees Fahrenheit. Notable figures are the high of 103 in the first decade and low of -2 also in the first decade. As well as, average temperatures spanning from 25.2 to 74.8.

Table 3: State College Temperature Summary by Season and 10-Year Period

Decade	Season	Avg High	Avg Low	Avg Temp	Hottest	Coldest	Days
06-15							
06-15	Winter	36.6	22.4	29.5	68	-9	902
06-15	Spring	59.6	40.0	49.8	93	0	920
06-15	Summer	79.9	61.7	70.8	101	41	920
06-15	Fall	61.1	44.1	52.6	91	13	910
16-25							
16-25	Winter	38.6	24.3	31.5	74	-5	903
16-25	Spring	59.3	40.7	50.0	89	10	920
16-25	Summer	80.5	62.7	71.6	94	42	920
16-25	Fall	62.6	44.7	53.7	90	15	910

6.2 State College table interpretation

This table showcases the four seasons and temperature data for State College across both decades in degrees Fahrenheit. Notable figures include the highest recorded temperature of 101°F during summer in the first decade and the lowest recorded temperature of -9°F during winter in the first decade. The average temperatures range from 29.5°F in winter during 2006-2015 to 71.6°F in summer during 2016-2025.

7 Plot

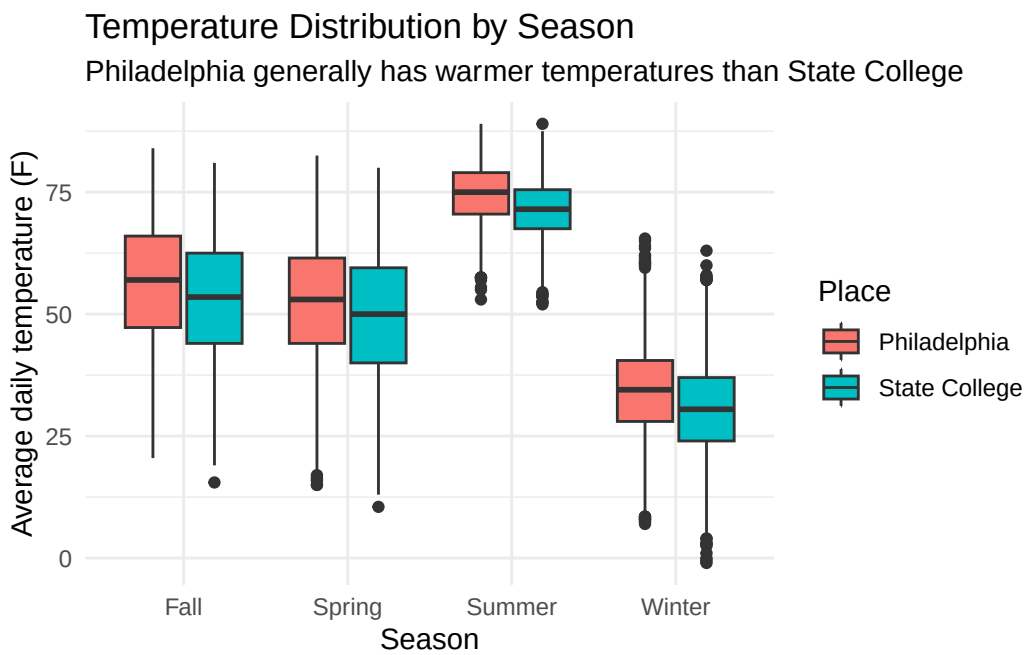

Figure 2: Temperature distribution by season for Philadelphia and State College

7.1 Analysis

Comparing the Philadelphia and State College tables and plot shows that Philadelphia is consistently warmer across every season and both 10-year periods. Philadelphia's average winter temperatures were 33.9°F from 2006-2015 and 35.0°F from 2016-2025, while State College's were lower at 29.5°F and 31.5°F. The same pattern appears in summer, where Philadelphia averaged 74.2°F and 74.8°F, compared to State College's 70.8°F and 71.6°F. Both locations show a slight increase in average temperatures from the first decade to the second, especially in winter and summer, suggesting a small warming trend over time. Philadelphia also had the highest recorded temperature overall at 103°F, while State College reached 101°F. However, State College had the colder extreme, dropping to -9°F, compared to Philadelphia's lowest temperature of -2°F. Overall, the tables suggest that Philadelphia has a warmer and slightly milder climate, while State College experiences colder winter conditions and a wider range of cold extremes. This could be credited to State College's higher elevation and distance from the coast.

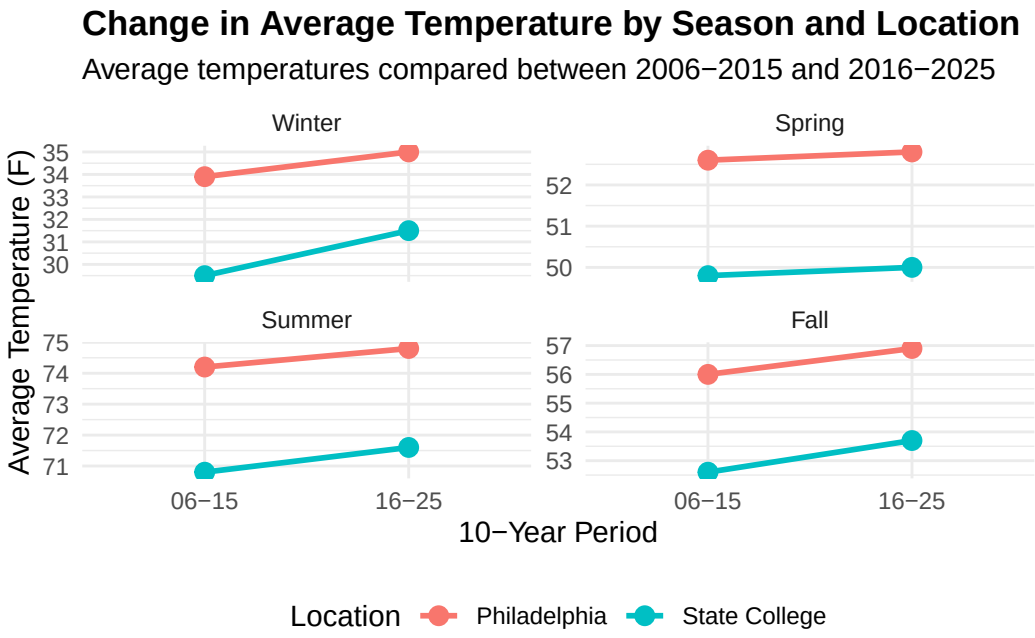

Figure 3: Change in average temperature between decades by season and location

7.2 Overall Trend

We can also see clearly from this plot that, while the changes are not massive, temperatures are rising in all four seasons and in both places. The slopes for all eight lines are very similar signifying that the temperature changes are at the least a regional trend. This could be caused by global warming or climate change.

8 Monthly Precipitation Trends

9 Narrative Text

This section compares the precipitation levels over a span of 20 years using NOAA datasets, focusing on the trends in Philadelphia and State College, Pennsylvania.

The visualization presents monthly precipitation totals for each city, with individual monthly observations shown as grey lines and the average trend line is blue. This approach allows the reader to see the variability per season as well as the long term changes in precipitation in both areas.

From the chart, both locations exhibit noticeable variability in monthly precipitation, with occasional spikes due to the changing seasons. We did notice a spike in rainfall in August 2011, more in Philadelphia than State College, but still a noticeable spike, but this was due to a couple of factors. The first major weather factor was Tropical Storm Lee, a slow moving storm that sat in one location dumping a lot of rain on already saturated ground and flooding the area. That happened in the beginning of August while Hurricane Irene showed up later in the month of August. A Category 1 storm, still brought substantial rainfall and giving little break for recovery between both larger systems. A low pressure system was also in place, pushing the storms out towards the Atlantic Ocean, which explains why the storms sat in Philadelphia longer than in State College. All of this being unfortunate timing combined for these historic rainfall conditions. Despite this one blip in the chart, the overall trends for both cities remain relatively similar despite their difference in elevation, which we belived would have an impact on the amount of rainfall between the cities.

Overall, the analysis demonstrates that while precipitation in Philadelphia and State College are broadly consistent over time, with both cities experiencing similar seasonal variability and long-term stability. Extreme events are outliers in this chart. Despite the geographic differences, both locations are generally similar and stable implying that regional weather systems have a stronger influence than just terrain, elevation, etc. This comparison provides a clear view of how precipitation behaves in the Mid-Atlantic region.

Monthly Precipitation Summary in Philadelphia vs State College

City	Avg (in)	Median (in)	Max (in)	Min (in)	Std Dev	Annual Avg (in)	Months Recorded
Philadelphia	3.93	3.62	15.03	0.00	2.14	47.15	244
State College	3.46	3.16	9.37	0.63	1.71	41.48	244

Figure 4: Precipitation summary table between Philadelphia and State College

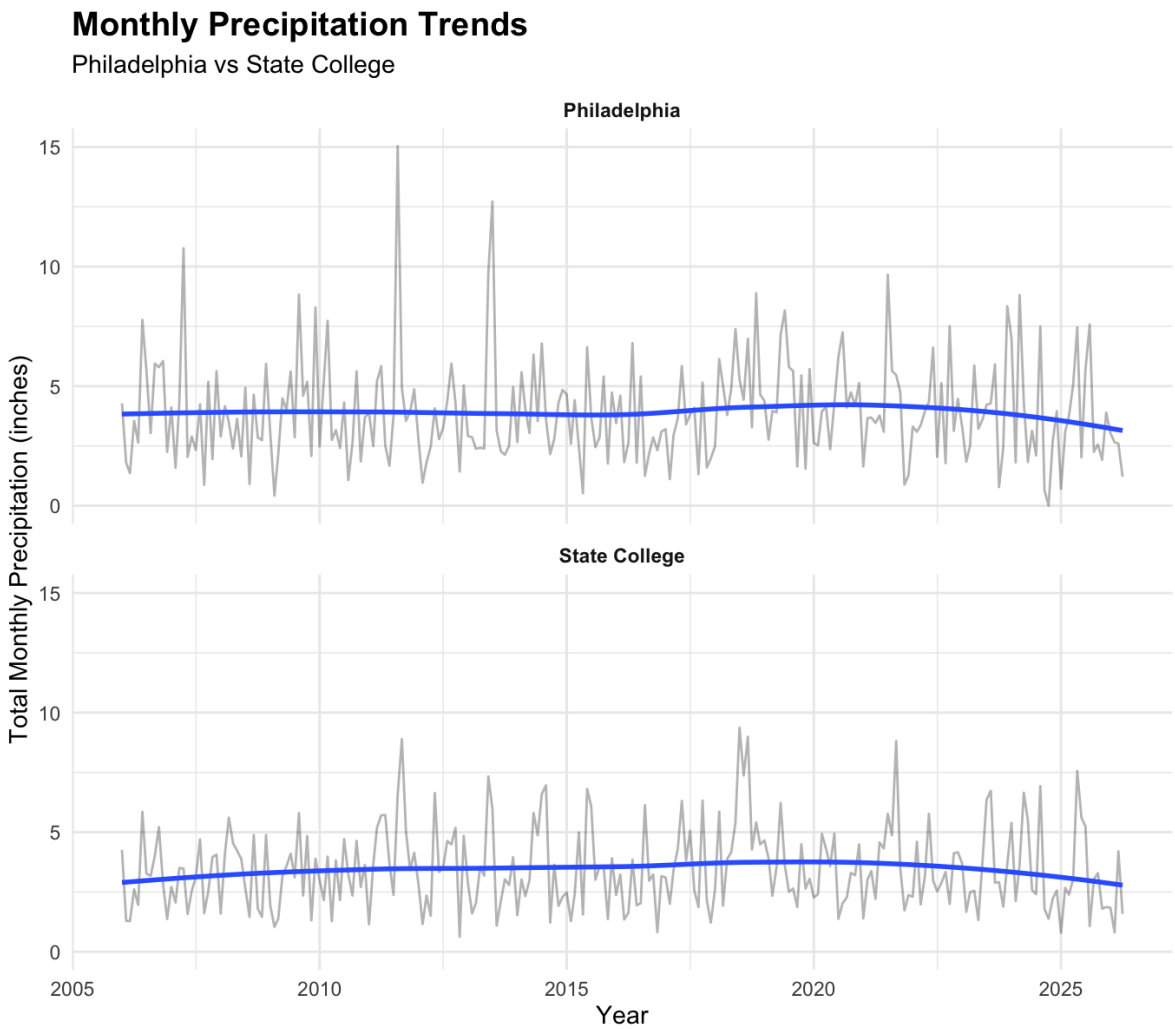

Figure 5: Precipitation levels month to month between Philadelphia and State College

10 Discussion

We don't technically need a dedicated section for this, we just need a larger narrative that connects our insights and explains the overall data story (Line 72 of the Project Guidelines File). The alternative would be writing a paragraph or two after the analysis of our final figure.

11 Author Contributions

Quinn - Paradigm/perspective - Data Provence - Temperature plots and insight

12 References

13 Code Appendix

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